Prof. Andrew Wilson
Department: Environmental Studies

Pre-requisite for a Summer Position: N/A

Description of Research:
Estimating the abundance of organisms is a basic element of ecology, field biology, and biological conservation. For songbirds, traditional survey methods include counting singing birds during the species' breeding season. In some instances--for example in terrain that is difficult to access--it can be challenging or time-consuming to conduct bird surveys on the ground. For the last few years, the Wilson lab has been pioneering the use of drones to conduct songbird surveys by attaching a lightweight recording device to a small quadcopter drone. Our previous research has shown that the technique is not only feasible but is potentially very time-efficient and can cause less disturbance to birds than traditional surveys where habitat is cross-crossing habitat on foot.

This year's research program will have three components:

1. compare bird counts from the drone system with counts from recorders placed on the ground
2. develop methods for fine-scale mapping of bird distribution
3. use a drone to produce fine-scale habitat maps that can be used to assess bird-habitat relationships

The fieldwork for all three parts will be conducted together as a team, but each of the three students will take the lead on field protocol development, data curation, and data analysis for one of the above. Students will need to be self-motivated to study for a remote drone pilot test during the first 2 to 3 weeks of the summer program
(https://www.faa.gov/uas/commercial_operators/become_a_drone_pilot)